

9

Using organometallic reagents to make C–C bonds

Connections

➡ Building on

- Electronegativity and the polarization of bonds [ch4](#)
- Grignard reagents and organolithiums attack carbonyl groups [ch6](#)
- C–H deprotonated by very strong bases [ch8](#)

Arriving at

- Organometallics: nucleophilic and often strongly basic
- Making organometallics from halo-compounds
- Making organometallics by deprotonating carbon atoms
- Using organometallics to make new C–C bonds from C=O groups

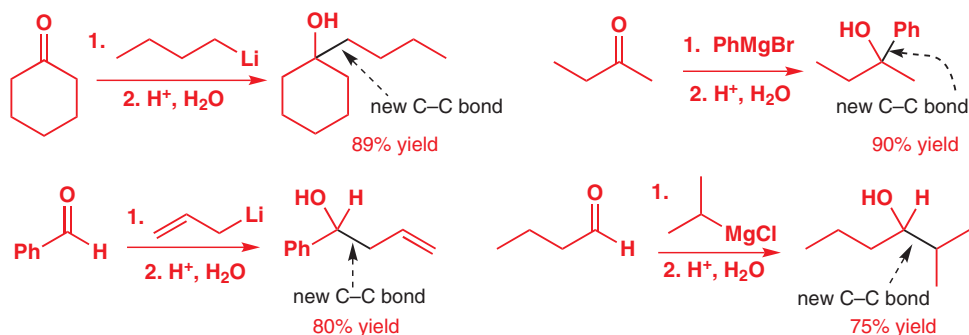
➡ Looking forward to

- More about organometallics [ch24](#) & [ch40](#)
- More ways to make C–C bonds from C=O groups [ch25](#), [ch26](#), & [ch27](#)
- Synthesis of molecules [ch28](#)

Introduction

In Chapters 2–8 we covered basic chemical concepts concerning *structure* (Chapters 2–4 and 7) and *reactivity* (Chapters 5, 6, and 8). These concepts are the bare bones supporting all of organic chemistry, and now we shall start to put flesh on these bare bones. In Chapters 9–22 we shall tell you about the most important classes of organic reaction in more detail.

One of the things organic chemists do, for all sorts of reasons, is to make molecules, and making organic molecules means making C–C bonds. In this chapter we are going to look at one of the most important ways of making C–C bonds: using organometallics, such as organolithiums and Grignard reagents, in combination with carbonyl compounds. We will consider reactions such as these:



You met these types of reactions in Chapter 6: in this chapter we will be adding more detail with regard to the nature of the organometallic reagents and what sort of molecules can be made using the reactions. The organometallic reagents act as nucleophiles towards the

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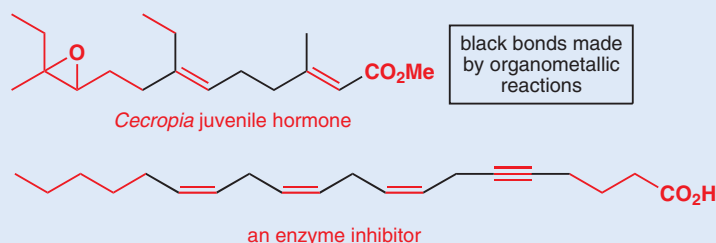
electrophilic carbonyl group, and this is the first thing we need to discuss: why are organometallics nucleophilic? We then move on to, firstly, how to make organometallics, then to the sorts of electrophiles they will react with, and finally to the sort of molecules we can make with them.

Organometallic compounds contain a carbon–metal bond

The polarity of a covalent bond between two different elements is determined by electronegativity. The more electronegative an element is, the more it attracts the electron density in the bond. So the greater the *difference* between the electronegativities, the greater the difference between the attraction for the bonding electrons, and the more polarized the bond becomes. In the extreme case of complete polarization, the covalent bond ceases to exist and is replaced by electrostatic attraction between ions of opposite charge. We discussed this in Chapter 4 (p. 96), where we considered the extreme cases of bonding in NaCl.

How important are organometallics for making C–C bonds?

As an example, let's take a molecule known as 'juvenile hormone'. It is a compound that prevents several species of insects from maturing and can be used as a means of controlling insect pests. Only very small amounts of the naturally occurring compound can be isolated from the insects, but it can instead be made in the laboratory from simple starting materials. At this stage you need not worry about how, but we can tell you that, in one synthesis, of the 16 C–C bonds in the final product, seven were made by reactions of organometallic reagents, many of them the sort of reactions we will describe in this chapter. This is not an isolated example. As further proof, take an important enzyme inhibitor, closely related to arachidonic acid which you met in Chapter 7. It has been made by a succession of C–C bond-forming reactions using organometallic reagents: eight of the 20 C–C bonds in the product were formed using organometallic reactions.

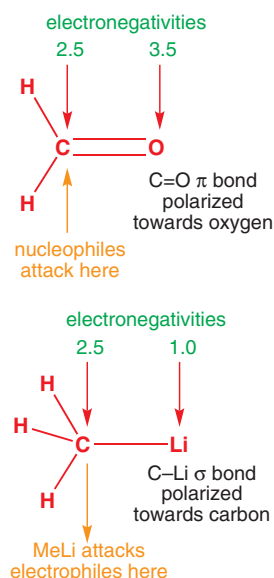


When we discussed (in Chapter 6) the electrophilic nature of carbonyl groups we saw that their reactivity is a direct consequence of the polarization of the carbon–oxygen bond towards the more electronegative oxygen, making the carbon a site for nucleophilic attack. In Chapter 6 you also met the two most important organometallic compounds—organolithiums and organomagnesium halides (known as Grignard reagents). In these organometallic reagents the key bond is polarized in the opposite direction—*towards* carbon—making carbon a nucleophilic centre. This is true for most organometallics because, as you can see from this edited version of the periodic table, metals (such as Li, Mg, Na, and Al) all have lower electronegativity than carbon.

Pauling electronegativities of selected elements

Li 1.0		B 2.0	C 2.5	N 3.0	O 3.5	F 4.0
Na 0.9	Mg 1.3	Al 1.6	Si 1.9	P 2.2	S 2.6	Cl 3.2

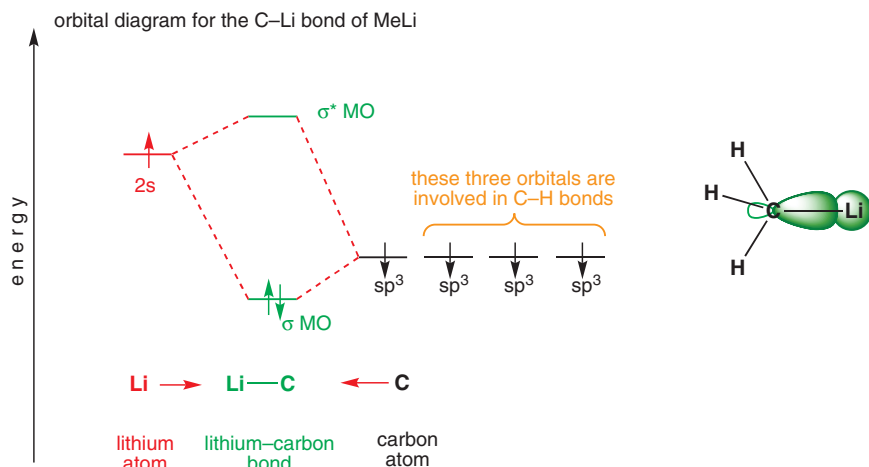
The molecular orbital energy level diagram—the kind you met in Chapter 4—represents the C–Li bond in methyllithium in terms of the sum of the atomic orbitals of carbon and lithium. The more electronegative an atom is, the lower in energy are its atomic orbitals (p. 96). The filled C–Li σ orbital is closer in energy to the carbon's sp^3 orbital than to the lithium's 2s orbital, so we can say that the carbon's sp^3 orbital makes a greater contribution to the C–Li σ bond and that the C–Li bond has a larger coefficient on carbon. Reactions involving the filled



Interactive display of polarity of organometallics

➡ We explained this reasoning on p. 104.

σ orbital will therefore take place at C rather than Li. The same arguments hold for the C–Mg bond of organo-magnesium or Grignard reagents, named after their inventor Victor Grignard.



■ Carbon atoms that carry a negative charge are known as **carbanions**.

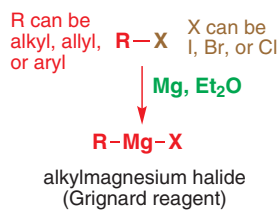
You have already met cyanide (p. 121), a carbanion that really does have a lone pair on carbon. Cyanide's lone pair is stabilized by being in a lower-energy sp orbital (rather than sp³) and by having the electronegative nitrogen atom triply bonded to the carbon.

We can also say that, because the carbon's sp³ orbital makes a greater contribution to the C–Li σ bond, the σ bond is close in structure to a filled C sp³ orbital—a lone pair on carbon. This useful idea can be carried too far: methyl lithium is not an ionic compound Me[–]Li⁺—although you may sometimes see MeLi or MeMgCl represented in mechanisms as Me[–].

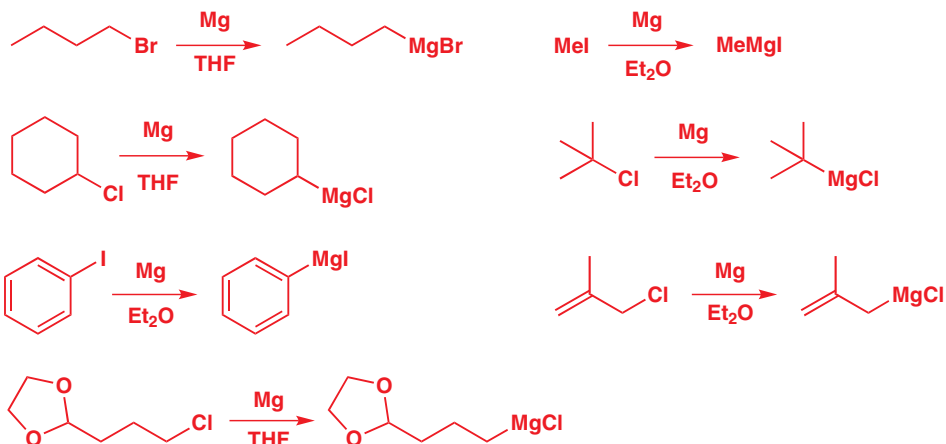
The true structure of organolithiums and Grignard reagents is rather more complicated! Even though these organometallic compounds are extremely reactive towards water and oxygen, and have to be handled under an atmosphere of nitrogen or argon, some have been studied by X-ray crystallography in the solid state and by NMR in solution. It turns out that they generally form complex aggregates with two, four, six, or more molecules bonded together, often with solvent molecules, one reason why apparently polar compounds such as BuLi dissolve in hydrocarbons. In this book we shall not be concerned with these details, and we shall represent organometallic compounds as simple monomeric structures.

Making organometallics

How to make Grignard reagents

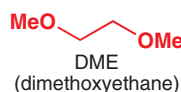
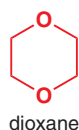
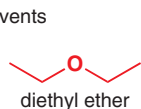


Grignard reagents are made by reacting magnesium turnings with alkyl halides in ether solvents to form solutions of alkylmagnesium halide. Iodides, bromides, and chlorides can be used, as can both aryl and alkyl halides. Our examples include methyl, primary, secondary, and tertiary alkyl halides, aryl and allyl halides. They cannot contain any functional groups that would react with the Grignard reagent once it is formed. The final example has an acetal functional group as an example of one that does not react with the Grignard reagent. (See Chapter 23 for further discussion.)



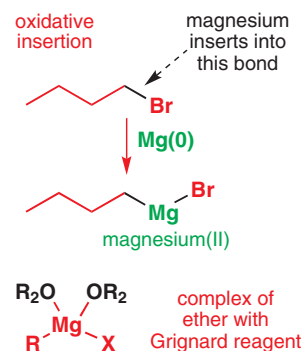
The solvents in these examples are all ethers, either diethyl ether Et_2O or THF. Other solvents that are sometimes used include the diethers dioxane and dimethoxyethane (DME).

common ether solvents



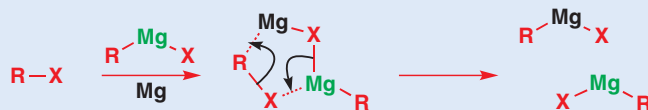
The reaction scheme is easy enough to draw, but what is the mechanism? Overall it involves an *insertion* of magnesium into the carbon–halogen bond. There is also a change in oxidation state of the magnesium, from $\text{Mg}(0)$ to $\text{Mg}(\text{II})$. The reaction is therefore known as an oxidative insertion or oxidative addition, and is a general process for many metals such as Mg , Li (which we meet shortly), Cu , and Zn . $\text{Mg}(\text{II})$ is much more stable than $\text{Mg}(0)$ and this drives the reaction.

The mechanism of the reaction is not completely understood, and probably involves radical intermediates. But what is sure is that by the end of the reaction the magnesium has surrendered its lone pair of electrons and gained two σ bonds. The true product is a complex between the Grignard reagent and, probably, two molecules of the ether solvent, as $\text{Mg}(\text{II})$ prefers a tetrahedral structure.



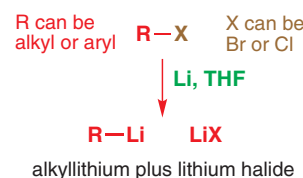
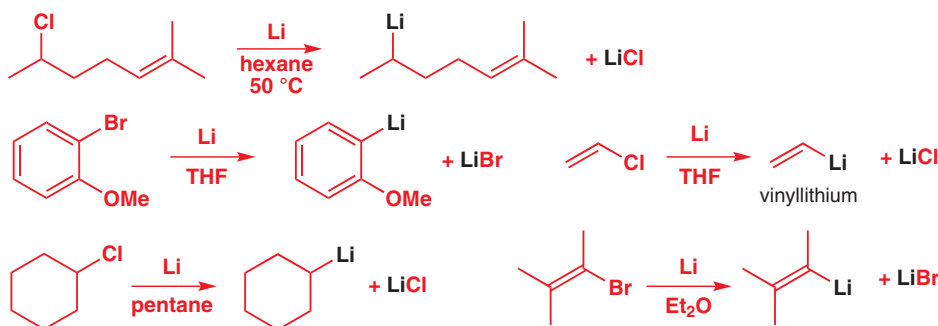
More on making Grignard reagents

The reaction takes place not in solution but on the surface of the metal, and how easy it is to make a Grignard reagent can depend on the state of the surface—how finely divided the metal is, for example. Magnesium is usually covered by a thin coating of magnesium oxide, and Grignard formation generally requires ‘initiation’ to allow the metal to come into direct contact with the alkyl halide. Initiation usually means adding a small amount of iodine or 1,2-diiodoethane, or using ultrasound to dislodge the oxide layer. Once the Grignard starts to form, it catalyses further reactions of $\text{Mg}(0)$, perhaps by this mechanism:



How to make organolithium reagents

Organolithium compounds may be made by a similar oxidative insertion reaction from lithium metal and alkyl halides. Each inserting reaction requires two atoms of lithium and generates one equivalent of lithium halide salt. As with Grignard formation, there is really very little limit on the types of organolithium that can be made this way.



You will notice secondary alkylolithiums, an aryllithium, and two vinylolithiums. The only other functional groups are alkenes and an ether. So far, that is quite like the formation of Grignard reagents. However, there are differences. Lithium goes from $\text{Li}(0)$ to $\text{Li}(\text{I})$ during the

Interactive mechanism for organolithium addition

reaction and there is no halide attached to the Li. Instead a second Li atom has to be used to make the Li halide. Again, Li(I) is very much more stable than Li(0) so the reaction is irreversible. Although ether solvents are often used, there is less need for extra coordination and hydrocarbon solvents such as pentane or hexane are also good.

Commercially available organometallics

Some Grignard and organolithium reagents are commercially available. Most chemists (unless they were working on a very large scale) would not usually make the simpler organolithiums or Grignard reagents by these methods, but would buy them in bottles from chemical companies (who, of course, do use these methods). The table lists some of the most important commercially available organolithiums and Grignard reagents.

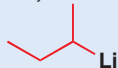
methyllithium (MeLi) in Et₂O or DME

***n*-butyllithium** (*n*-BuLi or just BuLi)



in cyclohexane or hexanes

sec-butyllithium (sec-BuLi or *s*-BuLi) in pentane or cyclohexane



tert-butyllithium (tert-BuLi or *t*-BuLi) in pentane



phenyllithium (PhLi) in (*n*-Bu)₂O

methylmagnesium chloride, bromide, and iodide (MeMgX) in Et₂O, or THF

ethylmagnesium bromide (EtMgBr)

butylmagnesium chloride (BuMgCl) in Et₂O or THF

allylmagnesium chloride and bromide



phenylmagnesium chloride and bromide (PhMgCl or PhMgBr) in Et₂O or THF

Organometallics as bases

Organometallics need to be kept absolutely free of moisture—even moisture in the air will destroy them. The reason is that they react very rapidly and highly exothermically with water to produce alkanes. Anything that can protonate them will do the same thing. The organometallic reagent is a strong base, and is protonated to form its conjugate acid—methane or benzene in these cases. The pK_a of methane (Chapter 8) is somewhere around 50: it isn't an acid at all and essentially nothing will remove a proton from methane.



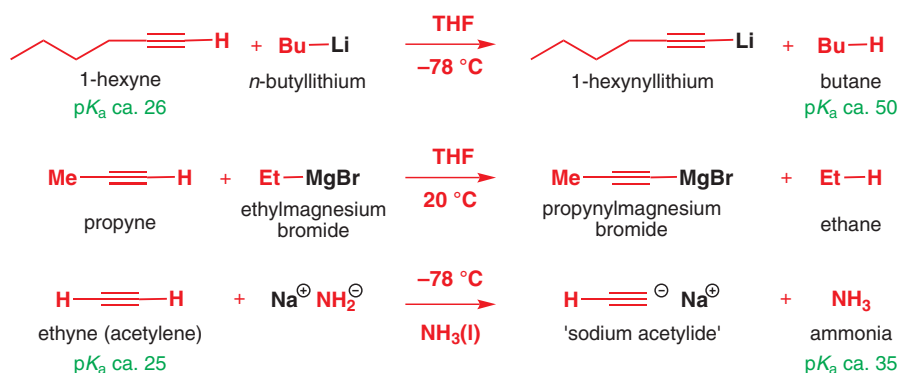
The equilibria lie vastly to the right: methane and Li⁺ are much more stable than MeLi while benzene and Mg²⁺ are much more stable than PhMgBr. Some of the most important uses of organolithiums—butyllithium, in particular—are as bases and, because they are so strong, they will deprotonate almost anything. That makes them very useful as reagents for making *other* organolithiums.



Making organometallics by deprotonating alkynes

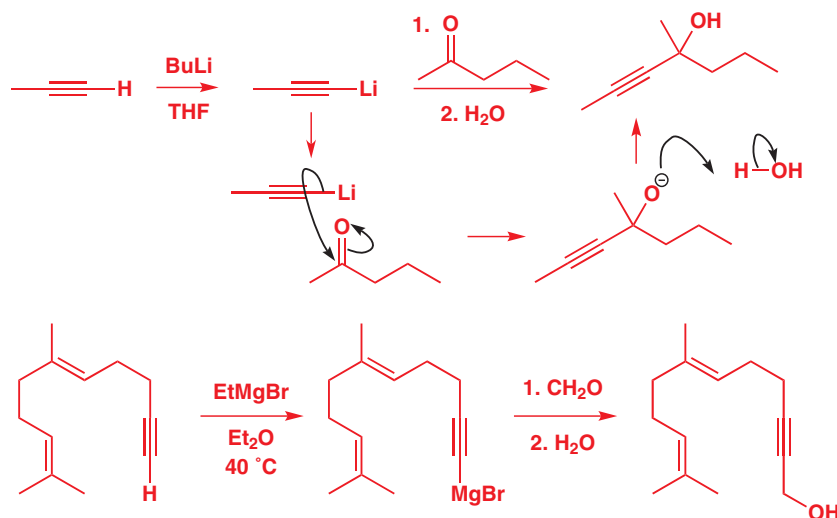
In Chapter 8 (p. 175) we talked about how hybridization affects acidity. Alkynes, with their C–H bonds formed from sp orbitals, are the most acidic of hydrocarbons, with pK_a s of about 25.

They can be deprotonated by more basic organometallics such as butyllithium or ethylmagnesium bromide. Alkynes are sufficiently acidic to be deprotonated even by nitrogen bases and you saw on p. 171 that a common way of deprotonating alkynes is to use NaNH_2 (sodium amide), obtained by reacting sodium with liquid ammonia. An example of each is shown here. Propyne and acetylene are gases, and can be bubbled through a solution of the base.



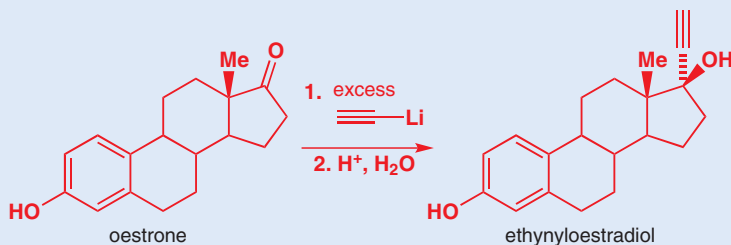
■ We have chosen to represent the alkynyl lithium and alkynyl magnesium halides as organometallics and the alkynyl sodium as an ionic salt. Both probably have some covalent character but lithium is less electropositive than sodium so alkynyl lithiums are more covalent and usually used in non-polar solvents while the sodium derivatives are more ionic and usually used in polar solvents.

The metal derivatives of alkynes can be added to carbonyl electrophiles, as in the following examples. The first (we have reminded you of the mechanism for this) is the initial step of an important synthesis of the antibiotic erythronolide A, and the second is the penultimate step of a synthesis of the widespread natural product farnesol.



Ethynylloestradiol

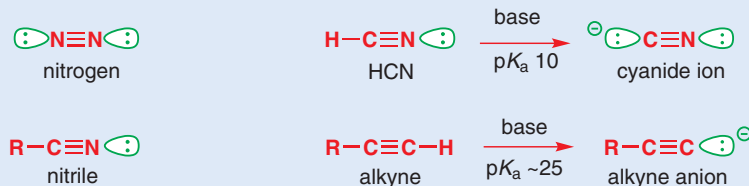
The ovulation-inhibiting component of almost all oral contraceptive pills is a compound known as ethynylloestradiol, and this compound too is made by an alkynyllithium addition to the female sex hormone oestrone. A range of similar synthetic analogues of hormones containing an ethynyl unit are used in contraceptives and in treatments for disorders of the hormonal system.



Triple bonds: stability and acidity

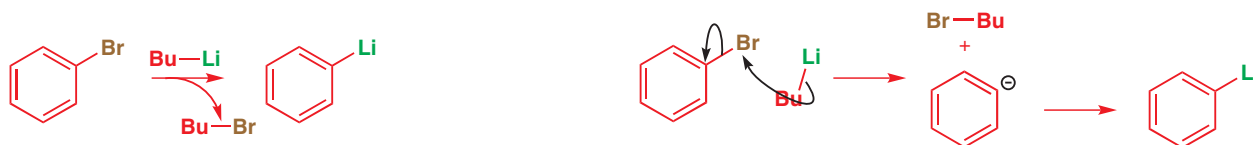
You have now met all the more important compounds with triple bonds. They all have electrons in low-energy sp hybrid orbitals (shown in green on the diagrams below), a feature which gives them stability or even unreactivity. Remember, an sp orbital has 50% s character, so electrons in this orbital are on average closer to the nucleus, and therefore more stable, than electrons in an sp^2 or sp^3 orbital.

Nitrogen, N_2 , has sp orbitals at both ends and is almost inert. It is neither basic nor nucleophilic and a major achievement of life is the 'fixing' (trapping in reductive chemical reactions) of nitrogen by bacteria such as those in the roots of leguminous plants (peas and beans). HCN has an sp orbital on nitrogen and a C-H σ bond at the other end. The nitrogen's sp lone pair is not at all basic, but HCN is quite acidic with a pK_a of 10 because the negative charge in the conjugate base (CN^-) is in an sp orbital. Nitriles have similar bonds and they are non-nucleophilic and non-basic. Finally, we have just met alkynes, which are among the most acidic of hydrocarbons, again because of the stability of an anion with its charge in an sp orbital.



Halogen-metal exchange

Deprotonation is not the only way to use one simple organometallic reagent to generate another more useful one. Organolithiums can also remove halogen atoms from alkyl and aryl halides in a reaction known as halogen-metal exchange.

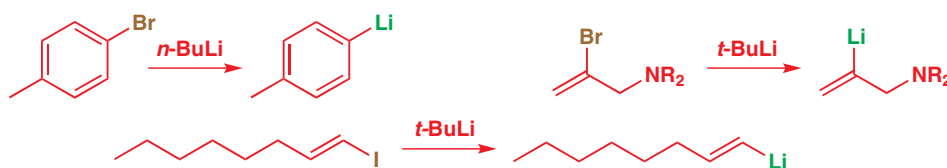


The bromine and the lithium simply swap places. As with many of these organometallic processes, the mechanism is not altogether clear, but can be represented as a nucleophilic attack on bromine by the butyllithium. But why does the reaction work? The product of our 'mechanism' is not PhLi and BuBr but a phenyl anion and a lithium cation. These could obviously combine to give PhLi and BuBr. But is this a reasonable interpretation and why does the reaction go that way and not the other? The key, again, is pK_a . We can think of the organolithiums as a complex between Li^+ and a carbanion.



■ The reason for this is again that the anion lies in an sp^2 orbital rather than an sp^3 orbital. See Chapter 8, p. 175.

The lithium cation is the same in all cases: only the carbanion varies. So the stability of the complex depends on the stability of the carbanion ligand. Benzene, (pK_a about 43) is more acidic than butane (pK_a about 50) so the phenyl complex is more stable than the butyl complex and the reaction is a way to make PhLi from available BuLi. Vinylolithiums (the lithium must be bonded directly to the alkene) can also be made this way and a R_2N- substituent is acceptable. Bromides or iodides react faster than chlorides.



Halogen–metal exchange tolls the knell of one appealing way to make carbon–carbon bonds. It may already have occurred to you that we might make a Grignard or organolithium reagent and combine it with another alkyl halide to make a new carbon–carbon σ bond.

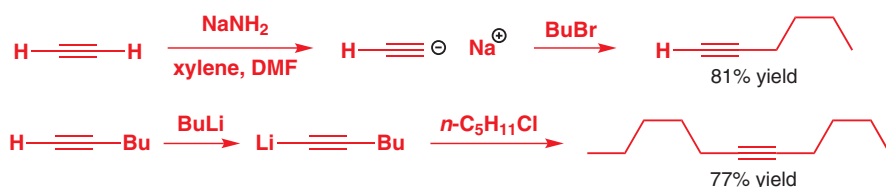


This reaction does not work because of transmetalation. The two alkyl bromides and their Grignard reagents will be in equilibrium with each other so that, even if the coupling were successful, three coupled products will be formed.



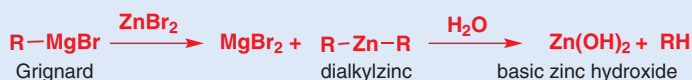
You will see later that transition metals are needed for this sort of reaction. The only successful reactions of this kind are couplings between metal derivatives of alkynes and alkyl halides. These do not exchange the metal as the alkynyl metal is much more stable than the alkyl metal.

A good example is the synthesis of a substituted alkyne starting from acetylene (ethyne) itself. One alkylation uses NaNH_2 as the base to make sodium acetylide and the other uses BuLi to make a lithium acetylide.

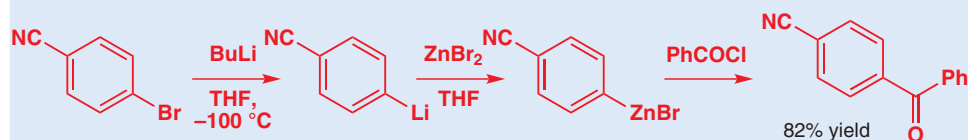


Transmetalation

Organolithiums can be converted to other types of organometallic reagents by transmetalation—simply treating with the salt of a less electropositive metal. The more electropositive Mg or Li goes into solution as an ionic salt, while the less electropositive metal such as Zn takes over the alkyl group.



But why bother? Well, the high reactivity—and in particular the basicity—of Grignard reagents and organolithiums sometimes causes unwanted side reactions. Their combination with very strong electrophiles like acid chlorides usually results in a violent uncontrolled reaction. If a much less reactive organozinc compound is used instead, the reaction is more under control. These organozinc compounds can be made from either Grignard reagents or organolithium compounds. E. Negishi, a pioneer of organozinc chemistry, got the Nobel Prize for Chemistry in 2010 with R. F. Heck and A. Suzuki for their work on organometallic compounds.



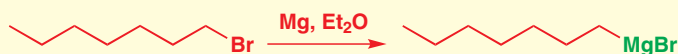
Using organometallics to make organic molecules

Now that you have met all of the most important ways of making organometallics (summarized here as a reminder), we shall move on to consider how to use them to make molecules:

what sorts of electrophiles do they react with and what sorts of products can we expect to get from their reactions? Having told you how you can make other organometallics, we shall really be concerned for the rest of this chapter only with Grignard reagents and organolithiums. In nearly all of the cases we shall talk about, the two classes of organometallics can be used interchangeably.

• Ways of making organometallics

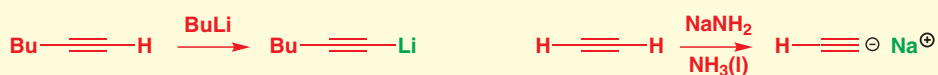
- Oxidative insertion of Mg into alkyl halides



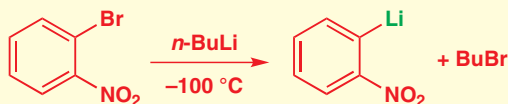
- Oxidative insertion of Li into alkyl halides



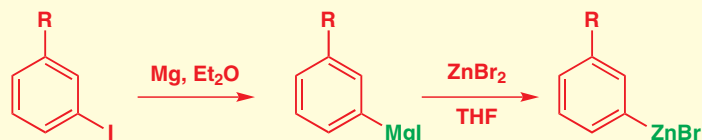
- Deprotonation of alkynes



- Halogen-metal exchange

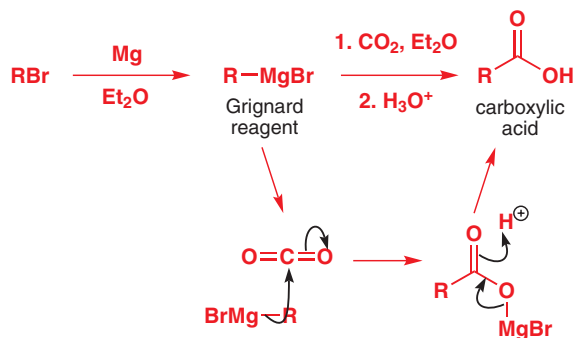


- Transmetalation



Making carboxylic acids from organometallics and carbon dioxide

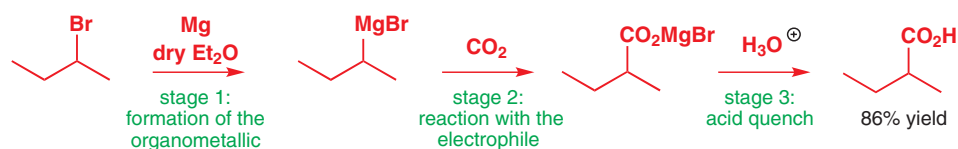
Carbon dioxide reacts with organolithiums and Grignard reagents to give carboxylate salts. Protonating the salt with acid gives a carboxylic acid with one more carbon than the starting organometallic. The reaction is usually done by adding solid CO_2 to a solution of the organolithium in THF or ether, but it can also be done using a stream of dry CO_2 gas.



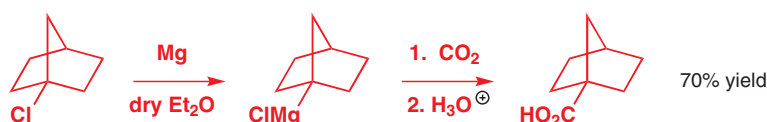
The example below shows the three stages of the reaction: (1) forming the organometallic, (2) reaction with the electrophile (CO_2), and (3) the acidic work-up or quench, which protonates the product and destroys any unreacted organometallic. The three stages of

the reaction have to be monitored carefully to make sure that each is finished before the next is begun. In particular it is absolutely essential that there is no water present during either of the first two stages—water must be added only at the end of the reaction, *after* the organometallic has all been consumed by reaction with the electrophile. You may occasionally see schemes written out without the quenching step included, but it is nonetheless always needed.

carboxylic acids from organometallics



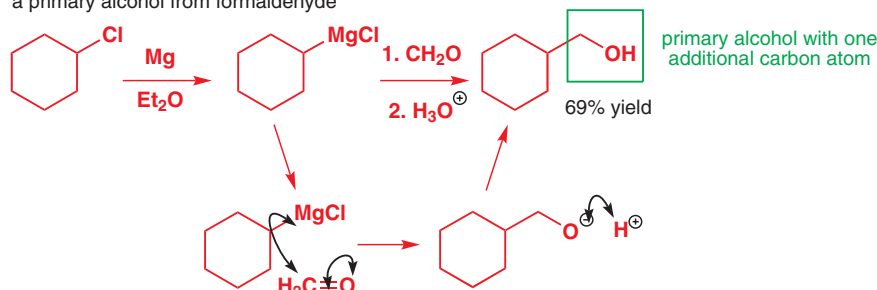
This next example shows that even very hindered chlorides can be used successfully. The significance of this will be clearer when you reach Chapter 15.



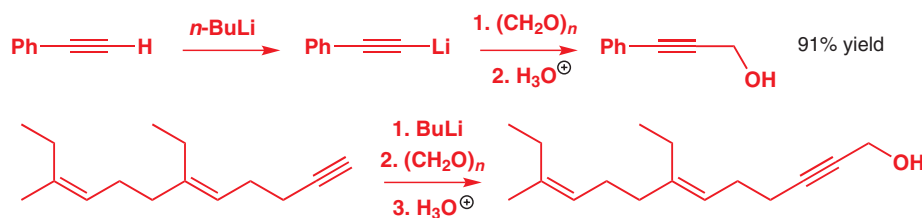
Making primary alcohols from organometallics and formaldehyde

You met formaldehyde, the simplest aldehyde, in Chapter 6, where we discussed the difficulties of using it in anhydrous reactions: it is either hydrated or a polymer paraformaldehyde, $(\text{CH}_2\text{O})_n$, and in order to get pure, dry formaldehyde it is necessary to heat ('crack') the polymer to decompose it. But formaldehyde is a remarkably useful reagent for making primary alcohols, in other words alcohols that have just one carbon substituent on the hydroxy-bearing C atom. Just as carbon dioxide adds one carbon and makes an acid, formaldehyde adds one carbon and makes an alcohol.

a primary alcohol from formaldehyde

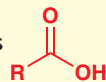


In the next two examples, formaldehyde makes a primary alcohol from two deprotonated alkynes. The second reaction here (for which we have shown organolithium formation, reaction, and quench simply as a series of three consecutive reagents) forms one of the last steps of the synthesis of *Cecropia* juvenile hormone, whose structure you met right at the beginning of the chapter.

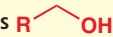


● **Something to bear in mind with all organometallic additions to carbonyl compounds is that the addition takes the oxidation level down one (oxidation levels were described in Chapter 2, p. 33). In other words, if you start with an aldehyde, you end up with an alcohol. More specifically,**

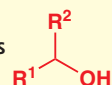
- additions to CO_2 give carboxylic acids



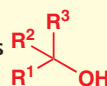
- additions to formaldehyde (CH_2O) give primary alcohols



- additions to other aldehydes (RCHO) give secondary alcohols



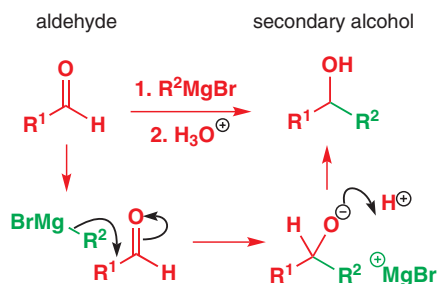
- additions to ketones give tertiary alcohols



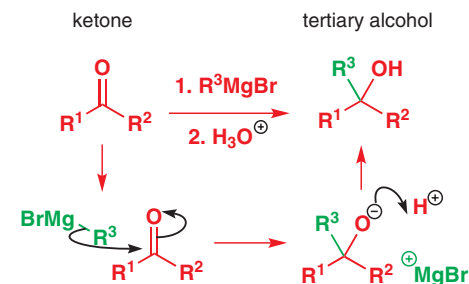
Secondary and tertiary alcohols: which organometallic, which aldehyde, which ketone?

Aldehydes and ketones react with organometallic reagents to form secondary and tertiary alcohols, respectively, and some examples are shown with the general schemes here.

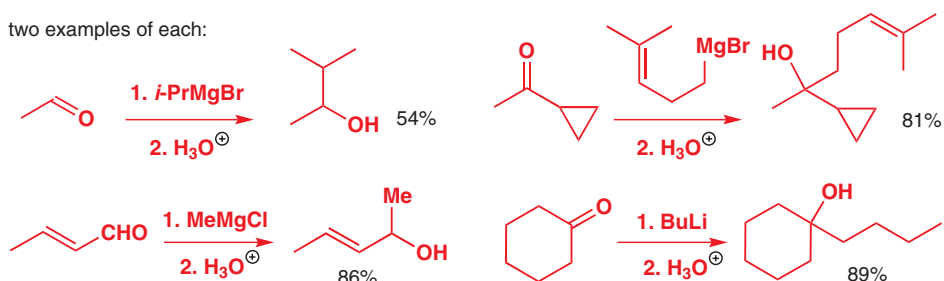
secondary alcohols from aldehydes



tertiary alcohols from ketones

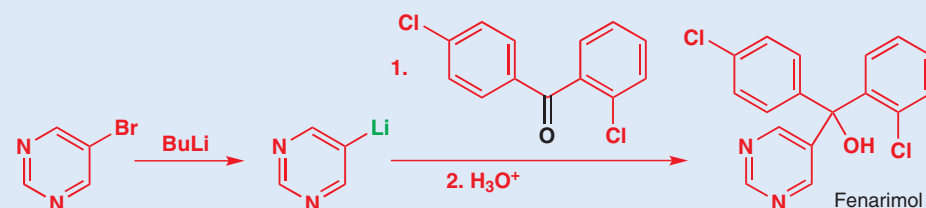


two examples of each:



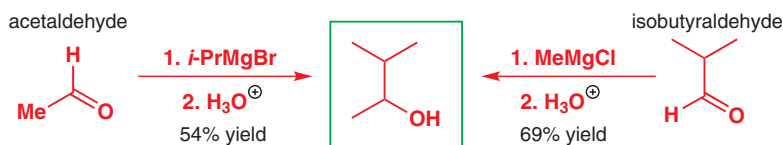
Fenarimol

Fenarimol is a fungicide that works by inhibiting the fungus's biosynthesis of important steroid molecules. It is made by reaction of a diarylketone with an organolithium derived by halogen–metal exchange.



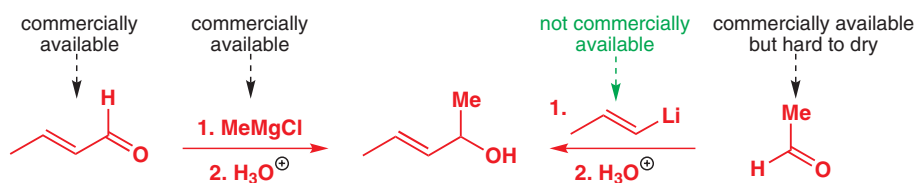
To make any secondary alcohol, however, there may be a choice of two possible routes, depending on which part of the molecule you choose to make the organometallic and which part you choose to make the aldehyde. For example, the first example here shows the synthesis of a secondary alcohol from isopropylmagnesium chloride and acetaldehyde. But it is

equally possible to make this same secondary alcohol from isobutyraldehyde and methyl-lithium or a methylmagnesium halide.



Indeed, back in 1912, when this alcohol was first described in detail, the chemists who made it chose to start with acetaldehyde, while in 1983, when it was needed as a starting material for a synthesis, it was made from isobutyraldehyde. Which way is better? The 1983 chemists probably chose the isobutyraldehyde route because it gave a better yield. But, if you were making a secondary alcohol for the first time, you might just have to try both in the laboratory and see which one gave a better yield.

Or you might be more concerned about which uses the cheaper, or more readily available, starting materials—this was probably also a factor in the choice of methylmagnesium chloride and the unsaturated aldehyde in the second example. Both can be bought commercially, while the alternative route to this secondary alcohol would require a vinyl lithium or vinylmagnesium bromide reagent that would have to be made from a vinyl halide, which is itself not commercially available, along with difficult-to-dry acetaldehyde.

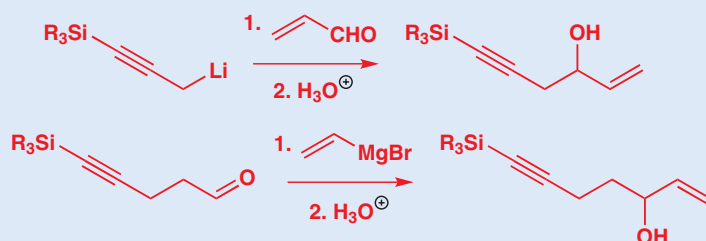


There is another choice for secondary alcohols: the reduction of a ketone. The ketone reacts with sodium borohydride to give a secondary alcohol. An obvious case where this would be a good route is the synthesis of a cyclic alcohol. This bicyclic ketone gives the secondary alcohol in good yield, and in the second example a diketone has both its carbonyl groups reduced.



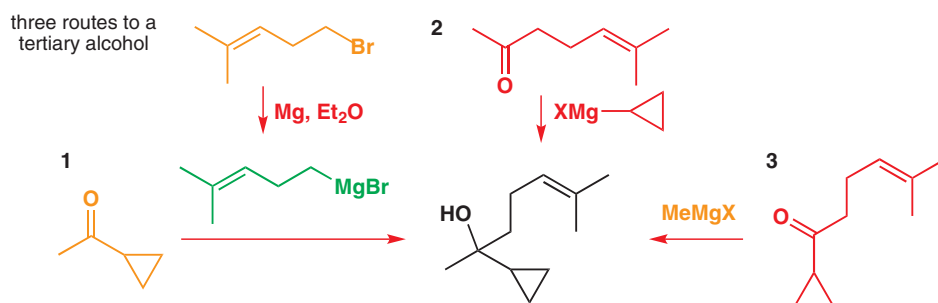
Flexibility in the synthesis of alcohols

As an illustration of the flexibility available in making secondary alcohols, one synthesis of bongkreic acid, a highly toxic compound that inhibits transport across certain membranes in the cell, requires both of these (very similar) alcohols. The chemists making the compound at Harvard University chose to make each alcohol from quite different starting materials: an unsaturated aldehyde and an alkyne-containing organolithium in the first instance, and an alkyne-containing aldehyde and vinyl magnesium bromide in the second.



With tertiary alcohols, there is even more choice. The example below is a step in a synthesis of the natural product, nerolidol. But the chemists in Paris who made this tertiary alcohol

could in principle have chosen any of three routes. Note that we have dropped the aqueous quench step from these schemes to avoid cluttering them.

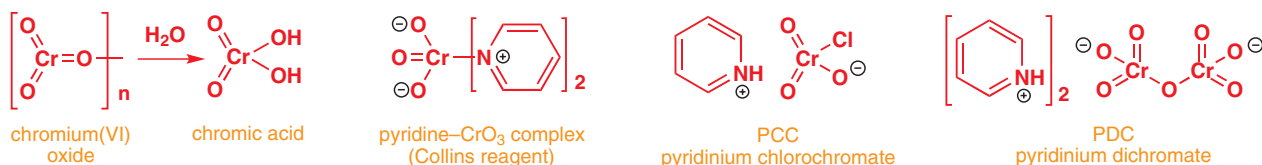


Only the reagents in orange are commercially available, but, as it happens, the green Grignard reagent can be made from an alkyl bromide, which is itself commercially available, making route 1 on the left the most reasonable.

Now, do not be dismayed! We are not expecting you to remember a chemical catalogue and to know which compounds you can buy and which you can't. All we want you to appreciate at this stage is that there are usually two or three ways of making any given secondary or tertiary alcohol, and you should be able to suggest alternative combinations of aldehyde or ketone and Grignard or organolithium reagent that will give the same product. You are not expected to be able to assess the relative merits of the different possible routes to a compound. That is a topic we leave for a much later chapter on retrosynthetic analysis, Chapter 28.

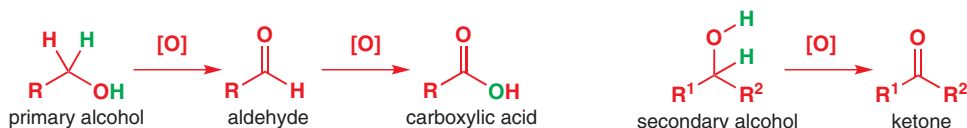
Oxidation of alcohols

So far the metals we have used have had one oxidation state other than zero: Li(I), Mg(II), and Zn(II). If we want to oxidize organic compounds we need metals that have at least two higher oxidation states and that means transition metals. The most important by far is chromium, with Cr(III) and Cr(VI) as the useful oxidation states. Orange Cr(VI) compounds are good oxidizing agents: they remove hydrogen from organic compounds and are themselves reduced to green Cr(III). There are many Cr(VI) reagents used in organic chemistry, some of the more important ones are related to the polymeric oxide CrO_3 . This is the anhydride of chromic acid and water breaks up the polymer to give a solution of chromic acid. Pyridine also breaks up the polymer to give a complex. This (Collins' reagent) was used to oxidize organic compounds but it is rather unstable and pyridinium dichromate (PDC) and pyridinium chlorochromate (PCC) are usually now preferred, especially as they are soluble in organic solvents such as CH_2Cl_2 .

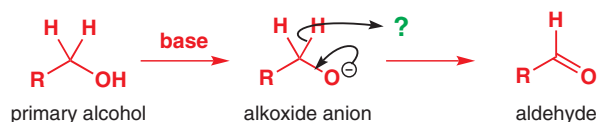


Oxidation by these reagents of the various primary and secondary alcohols we have been making in this chapter takes us to a higher oxidation level. Oxidation of primary alcohols gives aldehydes and then carboxylic acids, while oxidation of secondary alcohols gives ketones. Note that you can't oxidize tertiary alcohols (without breaking a C–C bond).

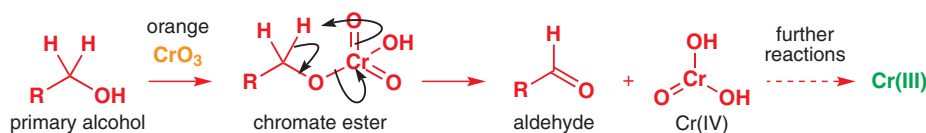
■ The symbol [O] means an unspecified oxidizing agent.



You will notice that the oxidation steps involve the removal of two hydrogen atoms and/or the addition of one oxygen atom. In Chapter 6 you saw that reduction meant the addition of hydrogen (and can also mean the removal of oxygen). Hiding behind these observations is the more fundamental idea that reduction requires the addition of electrons while oxidation requires the removal of electrons. If we used basic reagents, we could remove the OH proton from a primary alcohol, but to get the aldehyde we should have to remove a C–H proton as well *with a pair of electrons*. We should have to expel a hydride ion H^- and this doesn't happen. So we need some reagent that can remove a hydrogen atom *and a pair of electrons*. That defines an oxidizing agent.

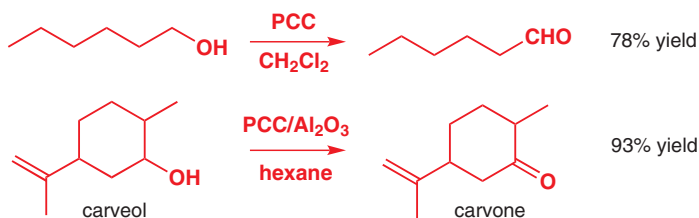


Here Cr(VI) can remove electrons to make Cr(III). It does so by a cyclic mechanism on a Cr(VI) ester. One hydrogen atom is removed (from the OH group) to make the ester and the second is removed (from carbon) in the cyclic mechanism. Notice how the arrows stop on the Cr atom and start again on the Cr=O bond, so two electrons are added to the chromium. This actually makes Cr(IV), an unstable oxidation state, but this gives green Cr(III) by further reactions.

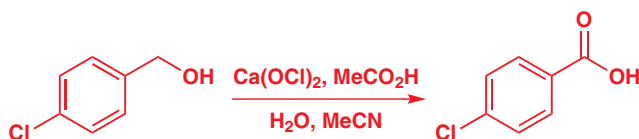


 Interactive mechanism for chromium (VI) oxidation of alcohols

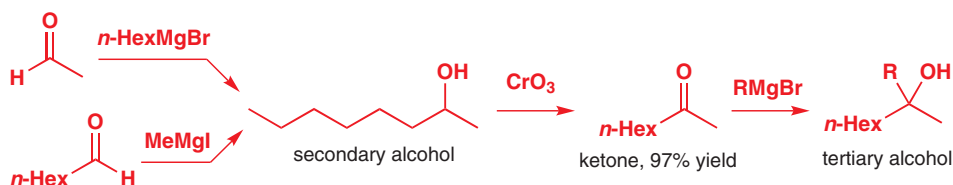
Two examples of the use of PCC in these oxidations come from Vogel. Hexanol is oxidized to hexanal in dichloromethane solution and commercial carveol (an impure natural product) to pure carvone with PCC supported on alumina in hexane solution. In both cases the pure aldehyde or ketone was isolated by distillation.



But a word of warning: stronger oxidizing agents like calcium hypochlorite or sodium hypochlorite (bleach) may oxidize primary alcohols all the way to carboxylic acids, especially in water. This is the case with *p*-chloro benzyl alcohol and the solid acid is easily isolated by the type of acid/base extraction we met in the previous chapter.



You will find further discussion of oxidizing agents in later chapters of the book. We have introduced them here so that you can see how primary and secondary alcohols, made by addition of organometallic reagents, can be oxidized to aldehydes or ketones so that the process can be repeated. A secondary alcohol, which could be made in two ways, can be oxidized with the pyridine–CrO₃ complex to the ketone and reacted with any Grignard or organolithium compound to give a range of tertiary alcohols.



Looking forward

In this chapter we have covered interconversions between ketones, aldehydes, and alcohols by forming C–C bonds using organometallics. We looked at oxidation and reduction as ways of complementing these methods—you should now be able to suggest at least one way of making any primary, secondary, or tertiary alcohol from simple precursors. In the next two chapters we will broaden our horizons beyond aldehydes and ketones to look at the reactivity of other carbonyl compounds—carboxylic acids and their derivatives such as esters and amides—and other nucleophiles. But the idea that we study organic reactions not only for their own sake but also so we can use them to make things should stay with you. We will come back to how to design ways of making molecules in Chapter 28. Many of these methods will employ the organometallics you have just met. We will then devote Chapter 40 to a broader range of more complex organometallic methods.

Further reading

For more on the detailed structures of Grignard reagents, see P. G. Williard in *Comprehensive Organic Synthesis*, vol. 1, 1999, p. 1. The alkylation of alkynes is described by P. J. Garratt in *Comprehensive Organic Synthesis*, vol. 1, 3rd edn, 1999, p. 271. The examples come from T. F. Rutledge, *J. Org. Chem.*, 1959, **24**, 840, D. N. Brattesoni and C. H. Heathcock, *Synth. Commun.* 1973, **3**, 245, R. Giovannini and P. Knochel, *J. Am. Chem. Soc.*, 1998, **120**, 11186, C. E. Tucker, T. N. Majid, and P. Knochel, *J. Am. Chem. Soc.*, 1992, **114**, 3983. For a rather advanced review of organozinc compounds, see P. Knochel, J. J. Almerna Perea, and P. Jones, *Tetrahedron*, 1998, **54**, 8275.

Discovery of pyridinium chlorochromate (PCC): G. Piancatelli, A. Scettri, and M. D'Auria, *Synthesis*, 1982, 245; H. S. Kasmai, S. G. Mischke, and T. J. Blake, *J. Org. Chem.*, 1995, **60**, 2267 and PDC: E. J. Corey and J. W. Suggs, *Tetrahedron Lett.*, 1975, 2647. Details of oxidation experiments: B. S. Furniss, A. J. Hannaford, P. W. G. Smith, and A. R. Tatchell, *Vogel's Textbook of Practical Organic Chemistry*, 5th edn, Longman, Harlow, 1989, pp. 590 and 610; J. C. Gilbert and S. F. Martin, *Experimental Organic Chemistry*, Harcourt, Fort Worth, 2002, p. 507.

Check your understanding



To check that you have mastered the concepts presented in this chapter, attempt the problems that are available in the book's Online Resource Centre at <http://www.oxfordtextbooks.co.uk/orc/clayden2e/>